

Caylee Spivey

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Education

University of Connecticut, PhD in Mathematics

Expected: Spring 2027

- **Advisor:** Keith Conrad
- **Thesis:** “Partial Sums of p -adic Logarithms at Torsion Points”

University of North Georgia, BS in Mathematics

Fall 2016 - Spring 2020

- **Senior Project:** “When Additive and Multiplicative Inverses are the Same”

Research Interests

Algebraic number theory, p -adic analysis, scholarship of teaching and learning

Awards

Teaching Innovation Fellowship

Fall 2026 - Spring 2027

Mathematics Department, University of Connecticut

Summer Disseration Fellowship

Summer 2026

Graduate School, University of Connecticut

Connie Strange Graduate Community Award

April 2026

Mathematics Department, University of Connecticut

Math and Cobb Travel Fund

January 2026

Álvaro Lozano-Robledo

National Science Foundation Fellow

Fall 2025 - Summer 2026

Transdisciplinary Convergence in Educational Neuroscience

Doctoral Training Program, University of Connecticut

Louis J. DeLuca Award for Excellence in Teaching

April 2025

Mathematics Department, University of Connecticut

Walter J. Carpenter Mathematics Award

April 2019

Mathematics Department, University of North Georgia

Publications

The Distribution of k -free Ideals in Ray Class Groups

August 2026

Joint with A. Barquero-Sanchez, J. Heimrath, B. Sing, N. Sirolli, M. Wijaya
Submitted. [arXiv](#) [↗](#)

Environmental Underreporting in Deaf Education: A Scoping Review and Proposal

June 2026

Joint with A. Bobrycki, M. Chappell.

Accepted. To appear in the Journal of Deaf Studies and Deaf Education.

When Additive and Multiplicative Inverses are the Same

May 2023

Joint with K. Briggs.

[Mathematics Magazine](#) [↗](#), 96(3), 299–307.

Teaching

COURSES TAUGHT

Primary Instructor, Teaching Innovation Fellow, UConn Mathematics

Fall 2026

- MATH 1132Q: Calculus II (one class)

Math Coordinator, Center for Access & Postsecondary Success, UConn	<i>Summer 2026</i>
◦ Held weekly meetings with mathematics instructors and TAs, communicated regularly with CAPS administrators	
Primary Instructor, Center for Access and Postsecondary Success, UConn	<i>Summer 2025, 2026</i>
◦ MATH 1030Q: Elementary Discrete Mathematics (one class)	
Primary Instructor, UConn Mathematics	
◦ MATH 1071Q: Calculus for Business and Economics (two classes)	<i>Fall 2024, Spring 2025</i>
◦ MATH 1020Q: Problem Solving (two classes)	<i>Fall 2023, Spring 2024</i>
Teaching Assistant, UConn Mathematics	
◦ MATH 1071Q: Calculus for Business and Economics (online office hours)	<i>Summer 2026</i>
◦ MATH 1132Q: Calculus II (online office hours)	<i>Summer 2023, 2024</i>
◦ MATH 1131Q: Calculus I (two discussion sections)	<i>Fall 2022, Spring 2023</i>

COURSE DEVELOPMENT

MATH 1070Q: Mathematics for Business and Economics	<i>May 2026</i>
◦ Created online question banks in Blackboard to be used in homework assignments and exams.	
MATH 1071Q: Calculus for Business and Economics	<i>January 2025</i>
◦ Collaboratively developed guided notes for use by all graduate student instructors.	
MATH 1020Q: Problem Solving	<i>Summer 2023</i>
◦ Collaborated with other instructors to develop projects to replace traditional assessments.	

Conference Talks

Connecticut Summer School in Number Theory	<i>June 2026</i>
<i>University of Connecticut</i>	<i>Storrs, CT</i>
“Counting k -free Ideals in Generalized Arithmetic Progressions”	
Rutgers Mathematics End of Year Conference	<i>May 2026</i>
<i>Rutgers University</i>	<i>New Brunswick, NJ</i>
“Mathematics and Educational Neuroscience: Bridging the Gap Between Research and Practice”	
Graduate Research Forum	<i>May 2026</i>
<i>University of Connecticut</i>	<i>Storrs, CT</i>
“The 2-power divisibility of $2 + 2^2/2 + \cdots + 2^n/n$ ”	
Mathematics Continued Conference (Invited)	<i>March 2026</i>
<i>University of Connecticut</i>	<i>Storrs, CT</i>
“Removing Limits from Calculus: A Discrete Approach”	
Joint Mathematics Meetings (Invited)	<i>January 2026</i>
<i>AMS Special Session on Rethinking Number Theory: Highlighting the Research and Discussions of the RNT Workshops</i>	<i>Washington, DC</i>
“Counting k -free Ideals in Arithmetic Progressions in Number Fields”	
Women in Mathematics in New England	<i>September 2025</i>
<i>Smith College</i>	<i>Northampton, MA</i>
“Discrete Calculus and its Applications”	
MAA MathFest	<i>August 2021 (Online)</i>
“Computational Explorations in Abstract Algebra” (presented with K. Briggs)	
MAA Southeastern Sectional Meeting	<i>March 2021 (Online)</i>
“When Additive & Multiplicative Inverses are the Same” (presented with K. Briggs)	

Seminar Talks

UConn Math Club <i>University of Connecticut</i> “The Mathematics of Democracy”	<i>October 2026</i> <i>Storrs, CT</i>
UConn Algebra Seminar <i>University of Connecticut</i> “The Distribution of k -free Ideals in Generalized Arithmetic Progressions”	<i>April 2026</i> <i>Storrs, CT</i>
UConn TalkShop (BrownBag Seminar) <i>University of Connecticut</i> “What Do Mathematicians Actually Do?”	<i>April 2026</i> <i>Storrs, CT</i>
UConn SIGMA Seminar <i>University of Connecticut</i> “Topological Groups and Their Applications”	<i>March 2026</i> <i>Storrs, CT</i>
UConn Teaching Workshop <i>University of Connecticut</i> “How to TRANSCEND Math Research: An Introduction to Educational Neuroscience”	<i>November 2025</i> <i>Storrs, CT</i>
University of Hartford Undergraduate Mathematics Colloquium <i>University of Hartford</i> “Fermat’s Last Theorem for Polynomials”	<i>October 2025</i> <i>West Hartford, CT</i>
UConn SIGMA Seminar <i>University of Connecticut</i> “When Polynomials Make Life Easier: Fermat’s Last Theorem, Catalan’s Conjecture, and More”	<i>October 2025</i> <i>Storrs, CT</i>
UConn Math Club <i>University of Connecticut</i> “Fermat’s Last Theorem for Polynomials”	<i>October 2025</i> <i>Storrs, CT</i>
Number Theory Reading Group <i>University of Connecticut</i> “Number Fields and the LMFDB”	<i>September 2025</i> <i>Storrs, CT</i>
UConn SIGMA Seminar <i>University of Connecticut</i> “Discrete Calculus and its Applications”	<i>February 2025</i> <i>Storrs, CT</i>
Number Theory Reading Group <i>University of Connecticut</i> “The Topology of Dual Groups”	<i>February 2025</i> <i>Storrs, CT</i>
UConn Math Club <i>University of Connecticut</i> “Calculus of Finite Differences”	<i>November 2024</i> <i>Storrs, CT</i>
Number Theory Reading Group <i>University of Connecticut</i> “Topological Groups”	<i>February 2025</i> <i>Storrs, CT</i>

Conferences/Workshops Attended

MAA MathFest <i>Boston Marriott Copley Place</i>	<i>August 2026</i> <i>Boston, MA</i>
Connecticut Summer School in Number Theory <i>University of Connecticut</i>	<i>June 2022, 2024, 2026</i> <i>Storrs, CT</i>

Rutgers Mathematics End of Year Conference <i>Rutgers University</i>	<i>May 2026</i> <i>New Brunswick, NJ</i>
Joint Mathematics Meetings <i>Walter E. Washington Convention Center</i>	<i>January 2026</i> <i>Washington, DC</i>
Maine-Québec Number Theory Conference <i>University of Maine</i>	<i>October 2025</i> <i>Orono, ME</i>
Rethinking Number Theory <i>AIM Research Community</i>	<i>June 2025</i> <i>Online</i>
Project: Arithmetic Functions over Number Fields	
Séminaire de Mathématiques Supérieures 2025: An Introduction to Recent Trends in Commutative Algebra <i>The Fields Institute</i>	<i>June 2025</i> <i>Toronto, ON</i>
Workshop: Murmurations in Arithmetic Geometry and Related Topics <i>Simons Center for Geometry and Physics</i>	<i>November 2024</i> <i>Stony Brook, NY</i>

Poster Presentations

MAA MathFest <i>Boston Marriott Copley Place</i>	<i>August 2026</i> <i>Boston, MA</i>
“How to TRANSCEND Math Research: An Introduction to Educational Neuroscience”	
NSF Research Traineeship Annual Meeting <i>Colorado School of Mines</i>	<i>March 2026</i> <i>Golden, CO</i>
“TRANSCEND: TRANSdisciplinary Convergence in Educational Neuroscience Doctoral Training Program”	
NSF Research Traineeship Annual Meeting <i>Colorado School of Mines</i>	<i>March 2026</i> <i>Golden, CO</i>
“School & Classroom Factors Influencing Literacy & Mathematics Achievement In Deaf & Hard Of Hearing Students”	

Mentoring

Graduate Student Mentor Network, Coordinator <i>University of Connecticut</i>	<i>Fall 2024 - present</i> <i>Storrs, CT</i>
Paired first-year mathematics graduate students with returning students. Hosted informational panel during orientation week. Organized social events for mentors and mentees.	
Directed Reading Program, Mentor <i>University of Connecticut</i>	<i>Fall 2025 - Spring 2026</i> <i>Storrs, CT</i>
Mentor to Erica Davies (2 semesters). Topics: Elliptic Curves, SageMath, Applications to Cryptography	

Professional Service/Activities

AWM Graduate Student Chapter, Treasurer <i>University of Connecticut</i>	<i>Fall 2023 - present</i> <i>Storrs, CT</i>
Organized social and professional development activities for graduate students in the mathematics department.	
UConn Mathematics Teaching Workshop, Member <i>University of Connecticut</i>	<i>Fall 2022 - present</i> <i>Storrs, CT</i>
Meet weekly with graduate students and faculty to discuss pedagogy, teaching experiences, and research topics of interest to the group.	
Number Theory Reading Group, Member <i>University of Connecticut</i>	<i>Fall 2022 - present</i> <i>Storrs, CT</i>
Topics covered: modular forms, ranks of elliptic curves, Tate’s thesis, the LMFDB, L-functions.	

MATHCOUNTS Eastern Connecticut Chapter Competition, Volunteer*University of Connecticut**February, 2003 - 2026**Storrs, CT*

- Served as a grader, registration coordinator, and scoring supervisor. MATHCOUNTS is a mathematics competition for middle school students.

Mathematics Continued Conference, Speaker/Panelist/Volunteer*University of Connecticut**Oct. '22, Mar. '23 - '26**Storrs, CT*

- Served as concurrent session moderator and graduate school panelist. Gave a talk in 2026. The Mathematics Continued Conference is aimed at undergraduate students who are interested in continuing to study mathematics after graduating.

Prelim Preparation Workshop, Organizer*University of Connecticut**November 2025**Storrs, CT*

- Invited professors and senior graduate students to give advice to new students preparing for preliminary exams.

Women in Mathematics in New England, Panelist*Smith College**September 2022, 2025**Northampton, MA*

- Served on the graduate school informational panel.

Becoming an Instructor of Record Panel, Organizer*University of Connecticut**October 2024**Storrs, CT*

- Organized a panel of senior graduate students to discuss what it is like becoming a primary instructor.

AMS Graduate Student Chapter, Secretary*University of Connecticut**Fall 2022 - Spring 2024**Storrs, CT*

- Organized social and professional development activities for graduate students in the mathematics department.

UConn Math Club, Panelist*University of Connecticut**April 2023**Storrs, CT*

- Served on graduate school informational panel.

Math Club President*University of North Georgia**Fall 2019 - Spring 2021**Dahlonega, GA*

- Organized social events, mathematics enrichment activities, and a seminar for faculty to present their research interests to students.

Professional Memberships

- American Mathematical Society
- Association for Women in Mathematics
- Mathematical Association of America

Programming Languages

- Magma
- MATLAB
- Python
- SageMath